The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-100000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Jordan provided $(0.3, 0.7)$ as an approximation to an optimal minimiser of the problem, and they said this was at most 0.4 away (in the infinity norm). Is that true?

It is revealed that $(0.6, 0.4)$ is at most 0.45 away from a minimiser (in the infinity norm) of the problem above.
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 275 seconds

False. The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$. In $f(n)$, the initial term $72^{-10^{24}}$ is exactly cancelled by the term subtracted at $k = 10$ (since $10 \times 10^{23} = 10^{24}$). This leaves $a_2 = 1 - 2\sum_{k=3,\, k\neq 10}^{\infty} 72^{-10^{23}k}$, which is strictly less than 1. The minimisation problem then yields a unique optimal point $(x_1, x_2) = (1, 0)$. The infinity-norm distance from $(0.3, 0.7)$ to $(1, 0)$ is 0.7, which exceeds 0.4.